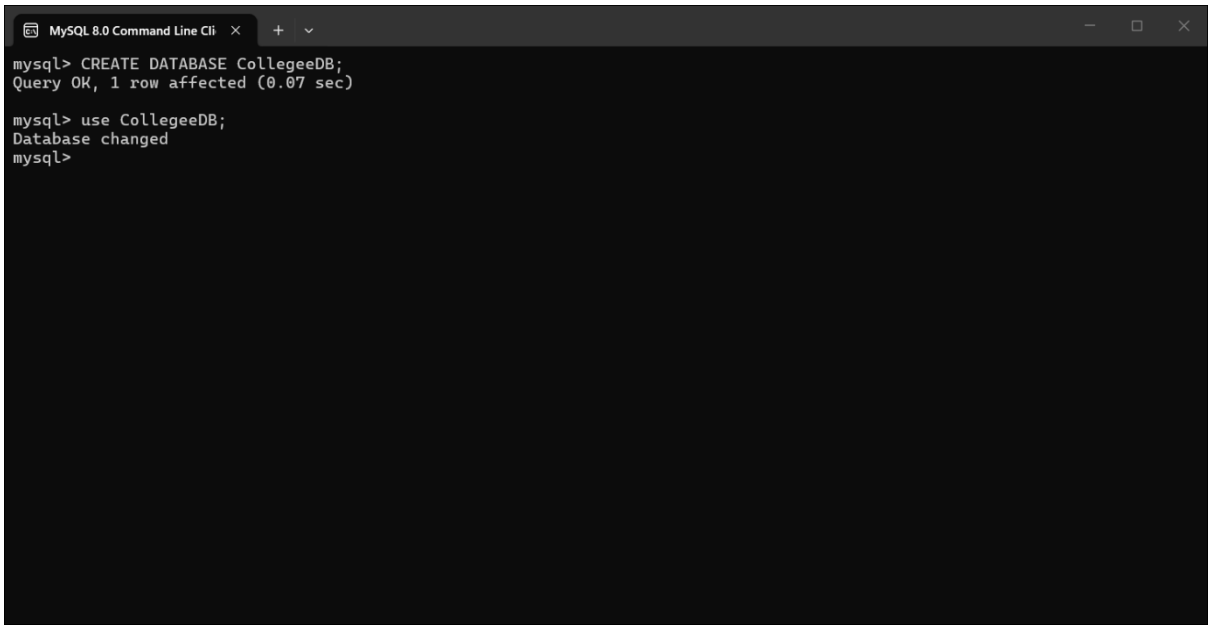
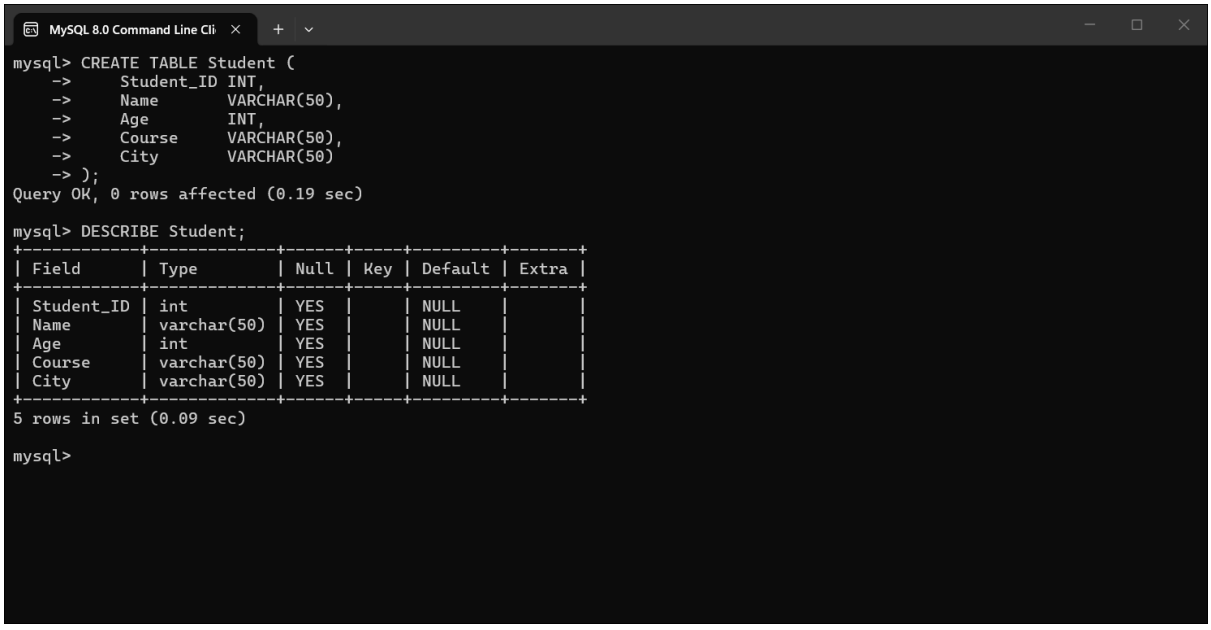
Lab Sheet 1

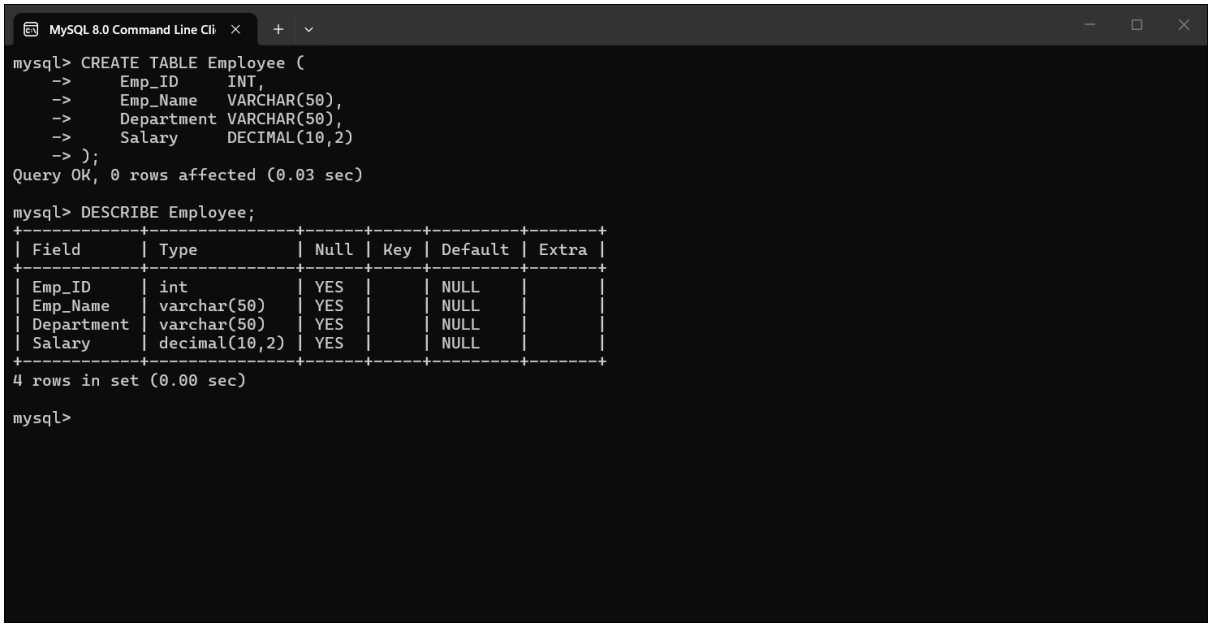
1. Create a database named CollegeDB.



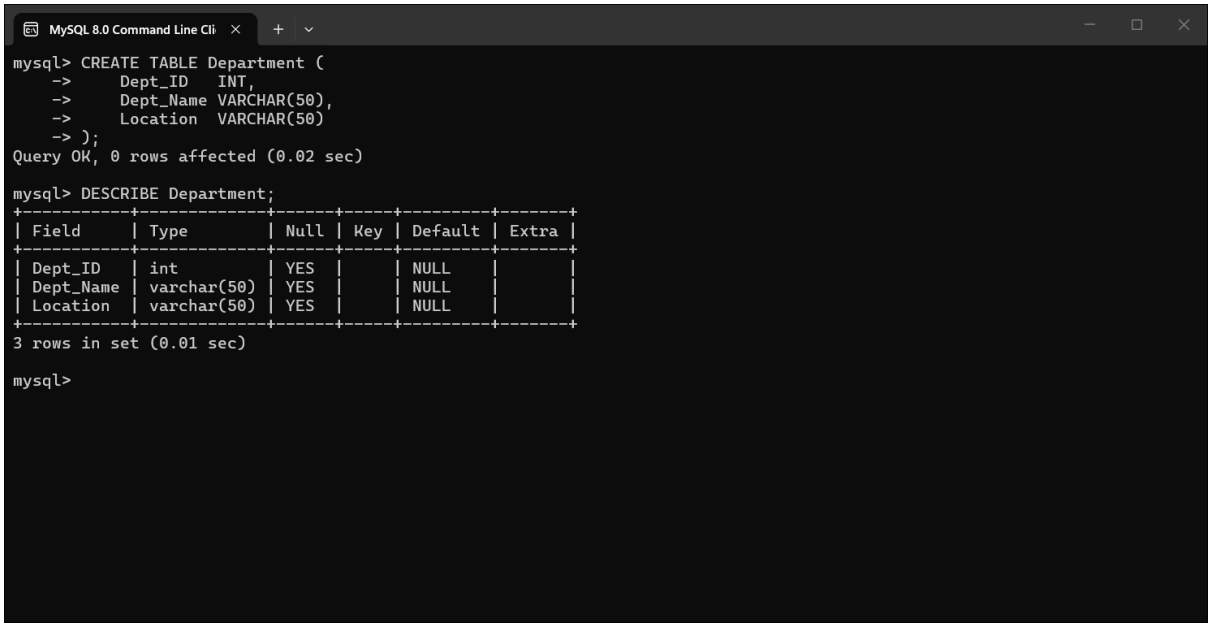
2. Create a Student table with Student_ID, Name, Age, Course and City.



3. Create an Employee table with Emp_ID, Emp_Name, Department and Salary.



4. Create a Department table with Dept_ID, Dept_Name and Location.



5. Create a Product table with Product_ID, Product_Name, Price and Quantity.

```
MySQL 8.0 Command Line Cli
mysql> CREATE TABLE Product (
    ->     Product_ID   INT,
    ->     Product_Name VARCHAR(100),
    ->     Price        DECIMAL(10,2),
    ->     Quantity     INT
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql> DESCRIBE Product;
+--------------+---------------+------+-----+---------+-------+
| Field        | Type          | Null | Key | Default | Extra |
+--------------+---------------+------+-----+---------+-------+
| Product_ID   | int           | YES  |     | NULL    |       |
| Product_Name | varchar(100)  | YES  |     | NULL    |       |
| Price        | decimal(10,2) | YES  |     | NULL    |       |
| Quantity     | int           | YES  |     | NULL    |       |
+--------------+---------------+------+-----+---------+-------+
4 rows in set (0.00 sec)

mysql>
```

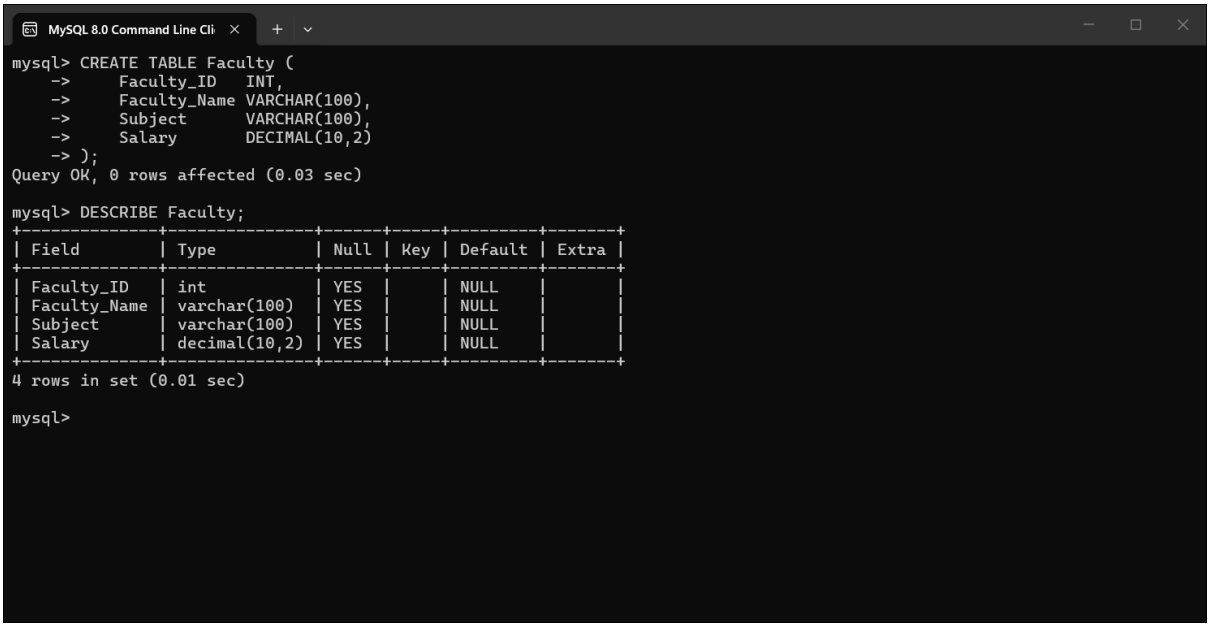
6. Create a Customer table with Customer_ID, Customer_Name, Email and Phone.

```
MySQL 8.0 Command Line Cli
mysql> CREATE TABLE Customer (
    ->     Customer_ID   INT,
    ->     Customer_Name VARCHAR(100),
    ->     Email         VARCHAR(100),
    ->     Phone         VARCHAR(15)
    -> );
Query OK, 0 rows affected (0.02 sec)

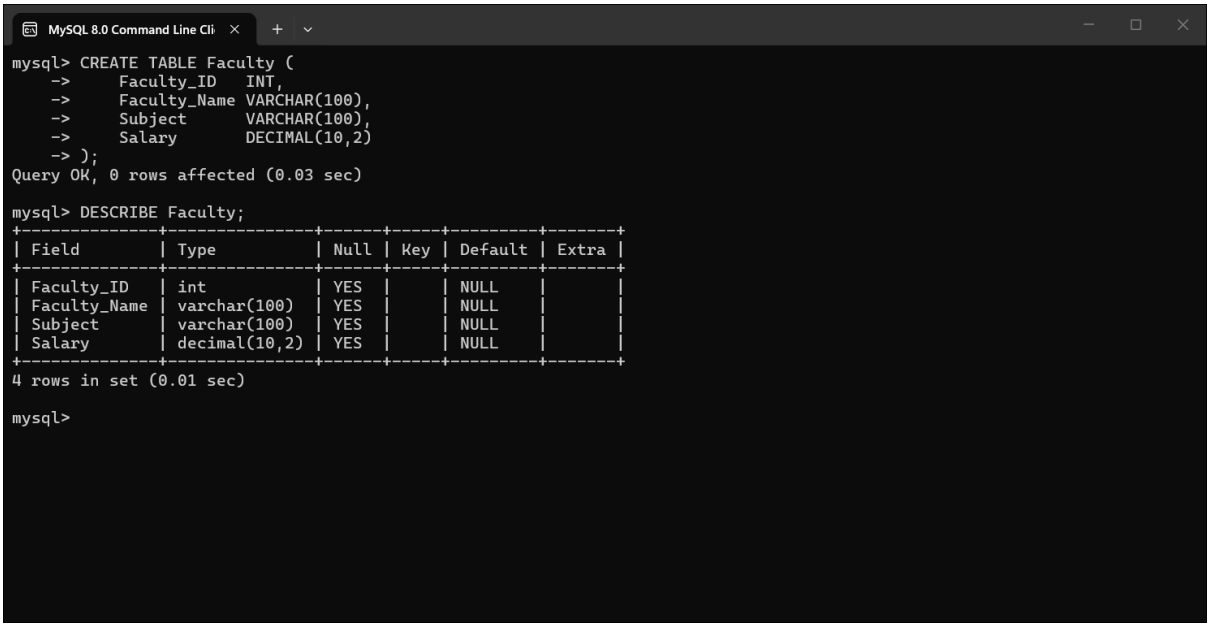
mysql> DESCRIBE Customer;
+---------------+--------------+------+-----+---------+-------+
| Field         | Type         | Null | Key | Default | Extra |
+---------------+--------------+------+-----+---------+-------+
| Customer_ID   | int          | YES  |     | NULL    |       |
| Customer_Name | varchar(100) | YES  |     | NULL    |       |
| Email         | varchar(100) | YES  |     | NULL    |       |
| Phone         | varchar(15)  | YES  |     | NULL    |       |
+---------------+--------------+------+-----+---------+-------+
4 rows in set (0.00 sec)

mysql>
```

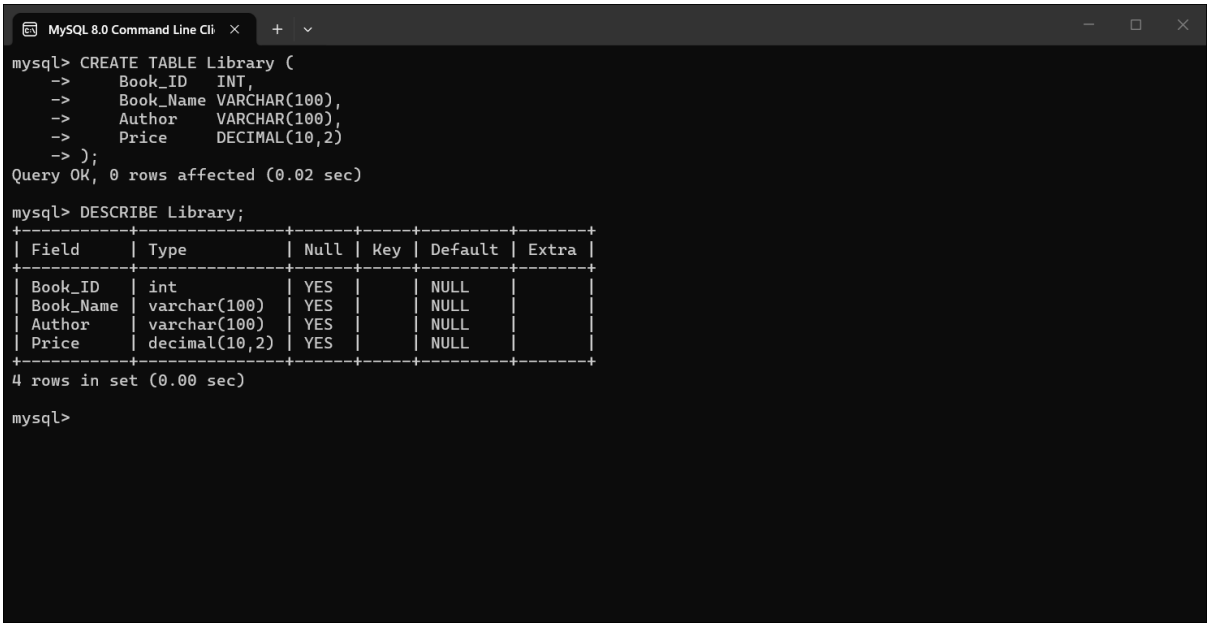
7. Create a Course table with Course_ID, Course_Name, Duration and Fees.



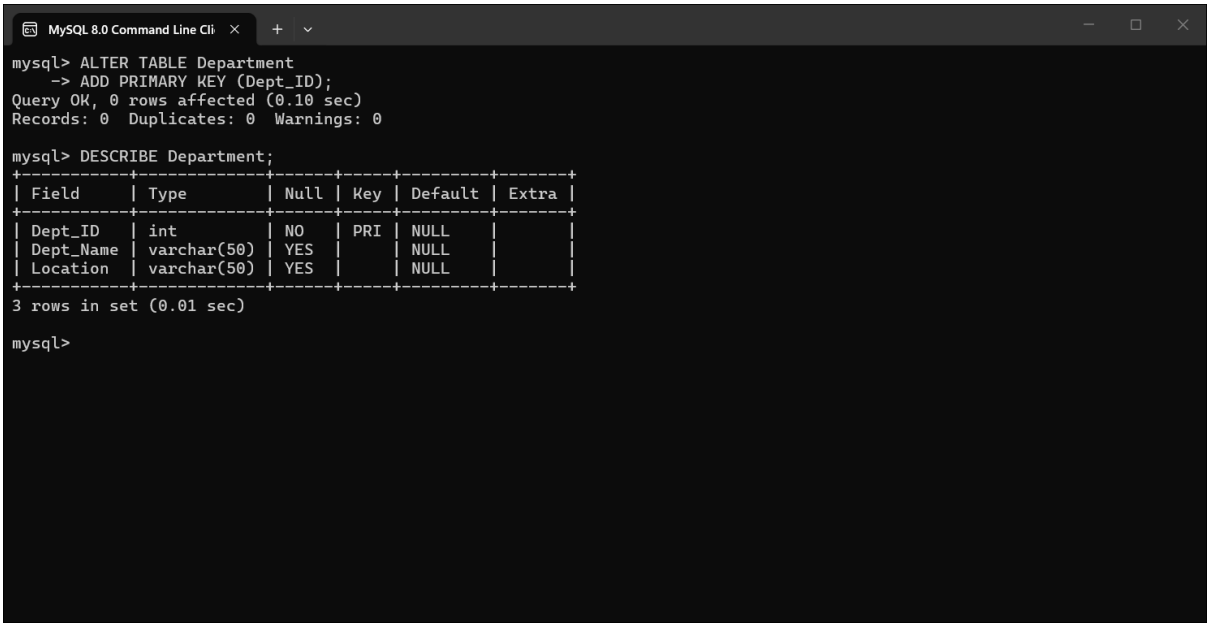
8. Create a Faculty table with Faculty_ID, Faculty_Name, Subject and Salary.



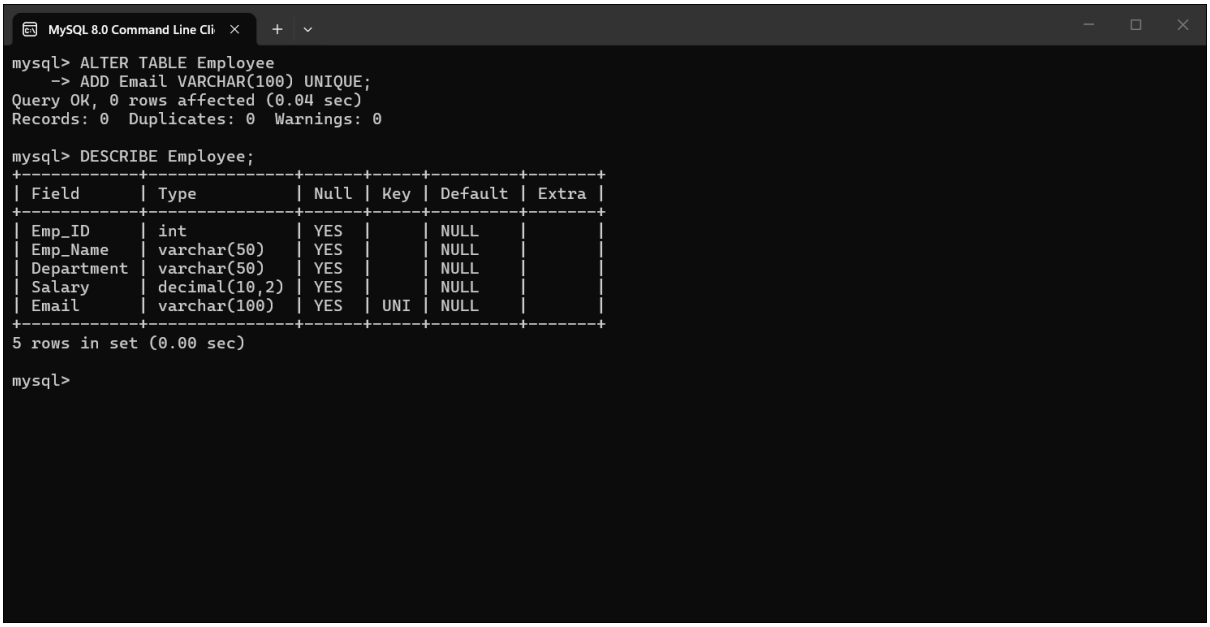
9. Create a Library table with Book_ID, Book_Name, Author and Price.



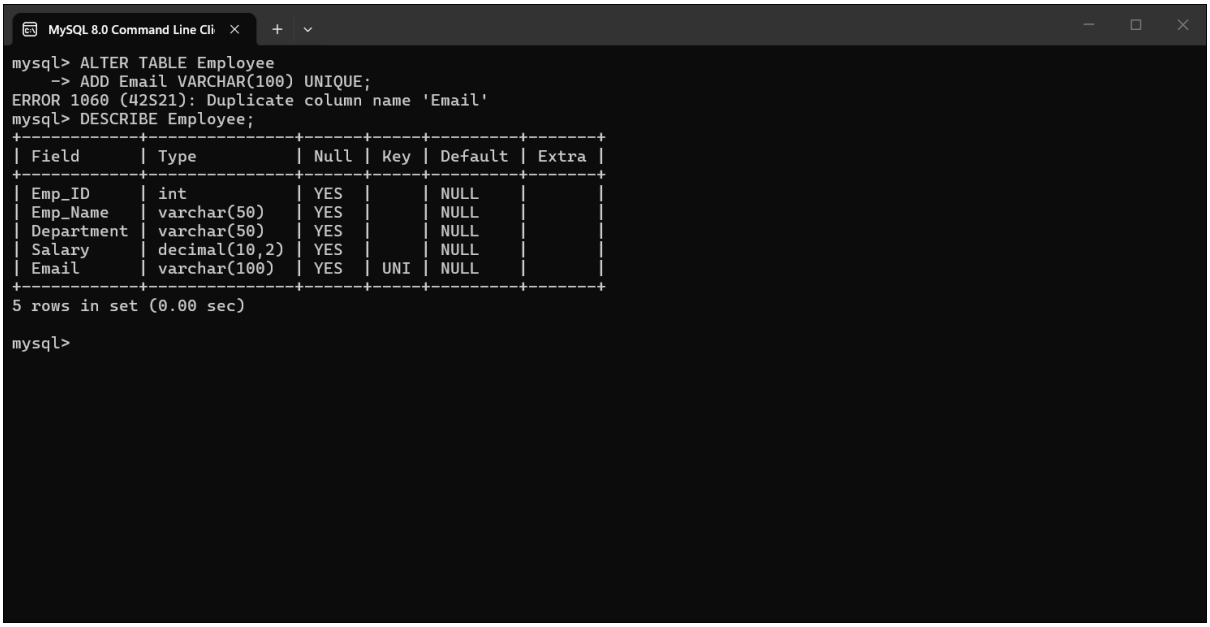
10. Create a Department table with Dept_ID as the Primary Key.



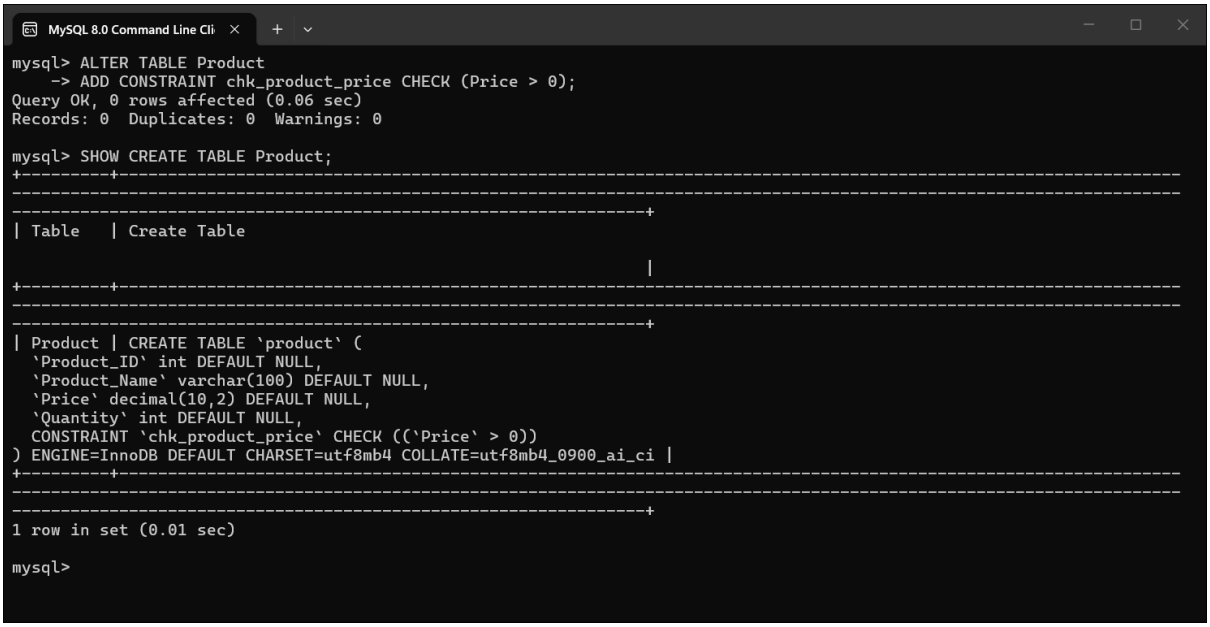
11. Create a Student table with Student_ID as Primary Key and Name as NOT NULL.



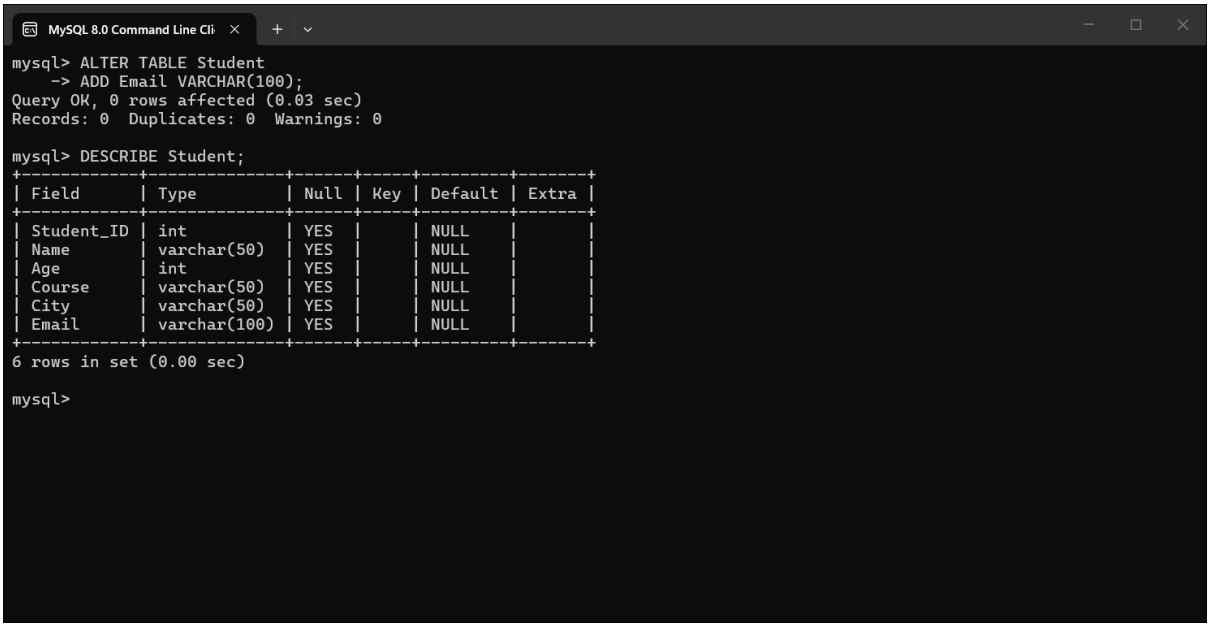
12. Create an Employee table with a UNIQUE constraint on Email



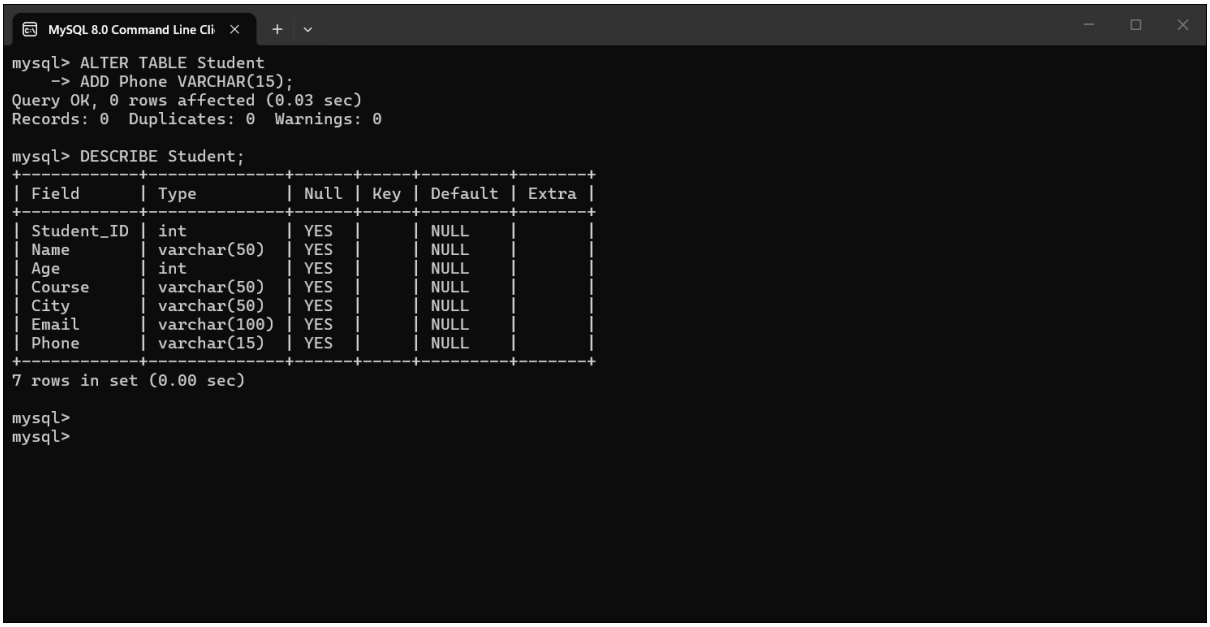
13. Create a Product table with a CHECK constraint that Price > 0.



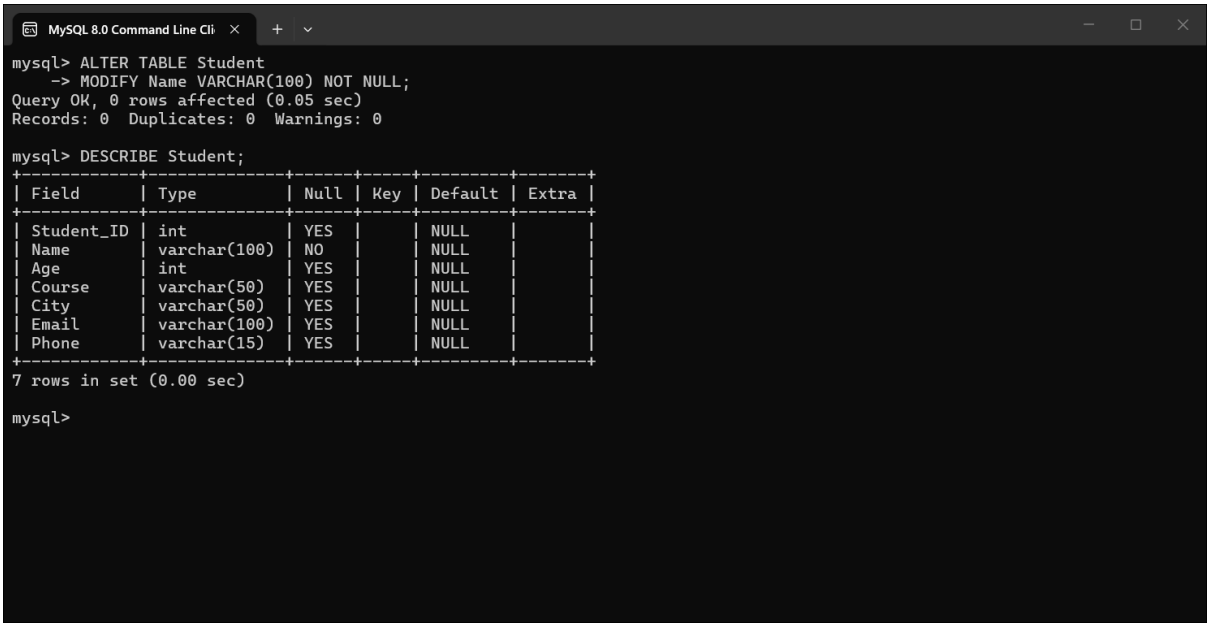
14. Add a new column Email to the Student table using ALTER TABLE.



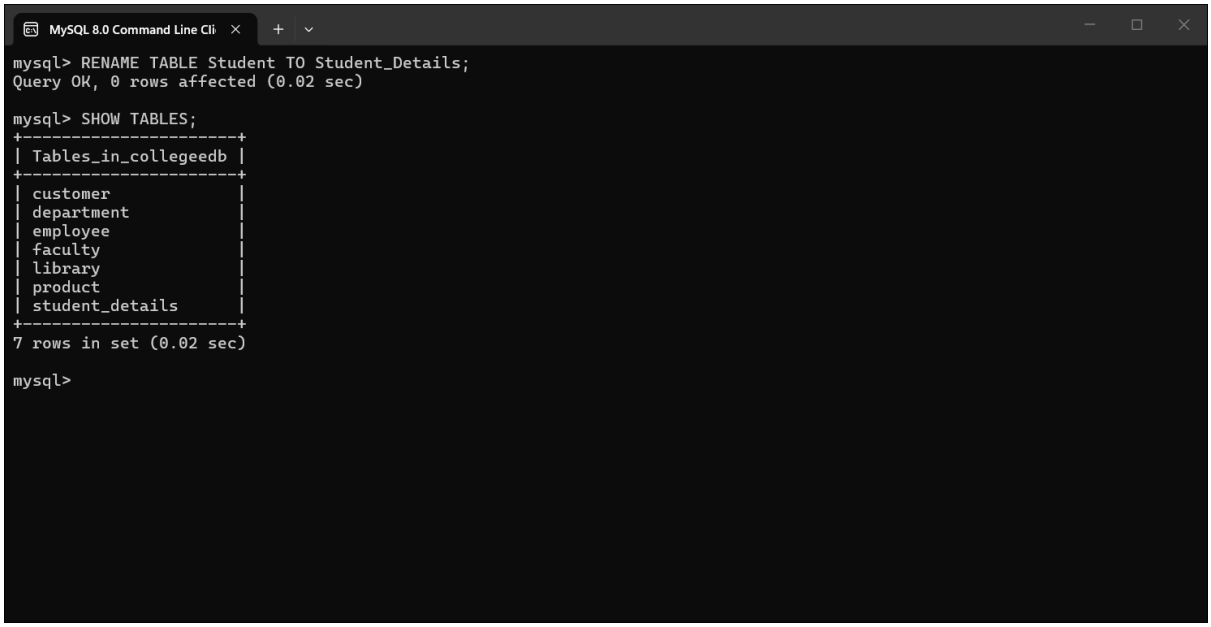
15. Add a Phone column to the Student table.



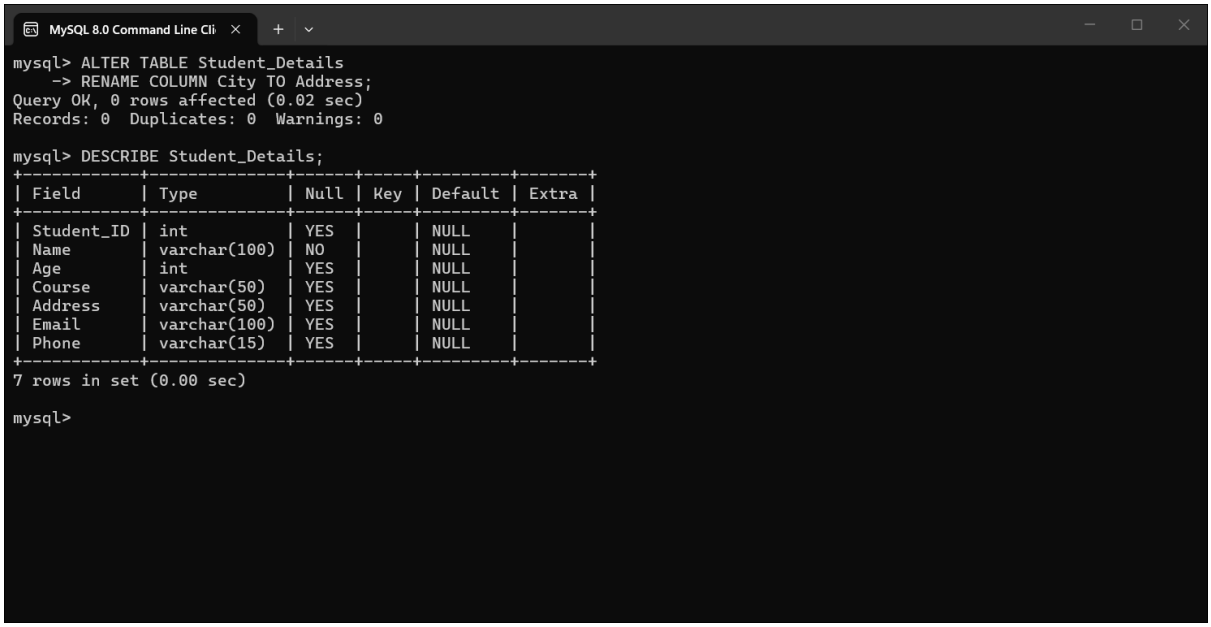
16. Modify the datatype/size of the Name column in the Student table.



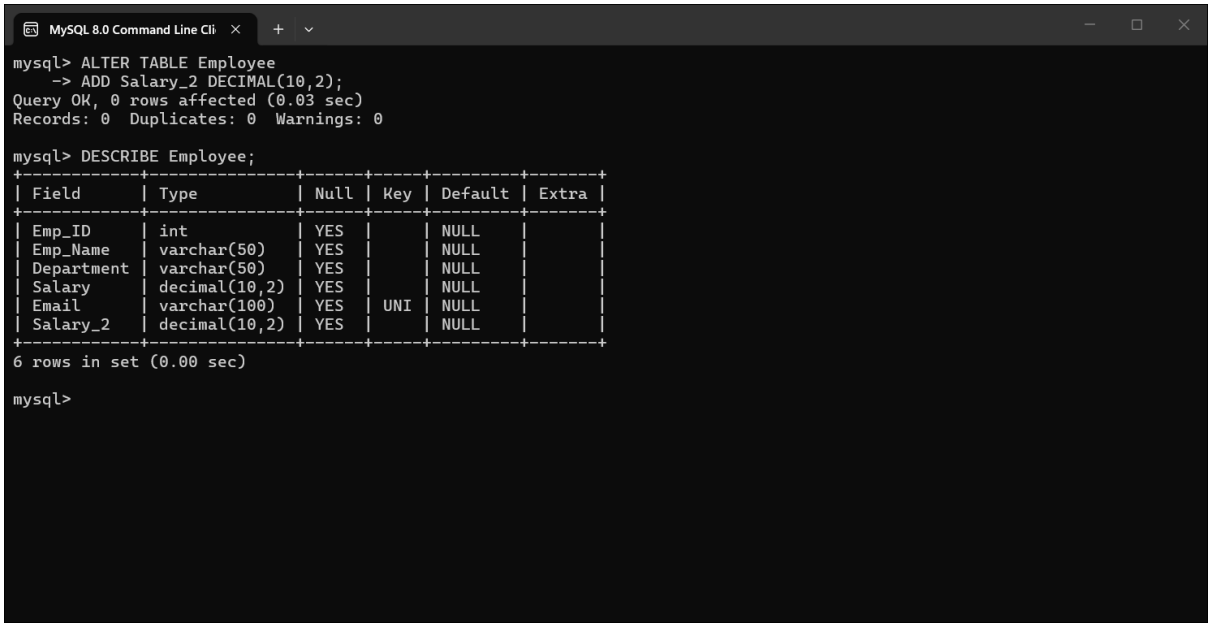
17. Rename the Student table to Student_Details.



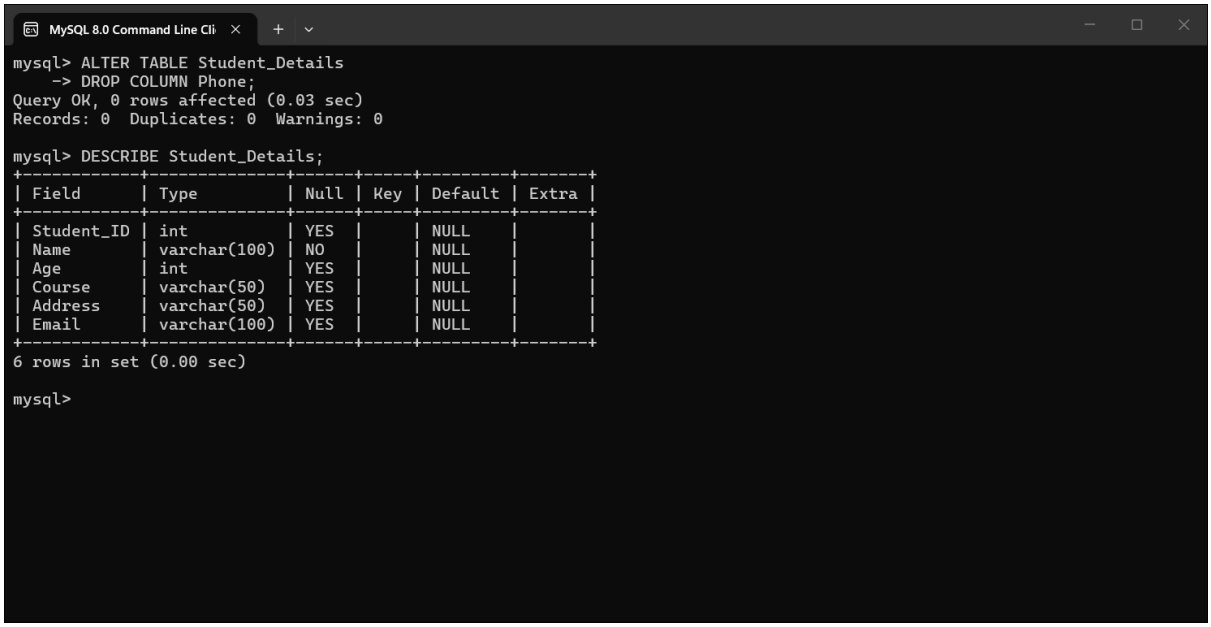
18. Rename the column City to Address.



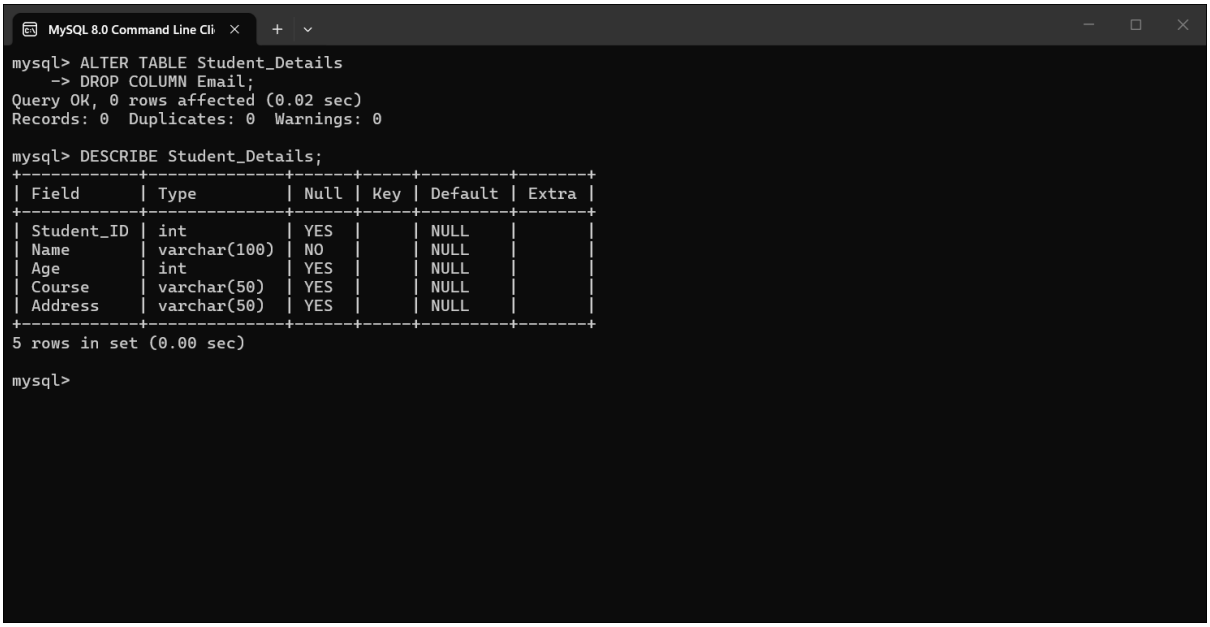
19. Add a Salary column to the Employee table.



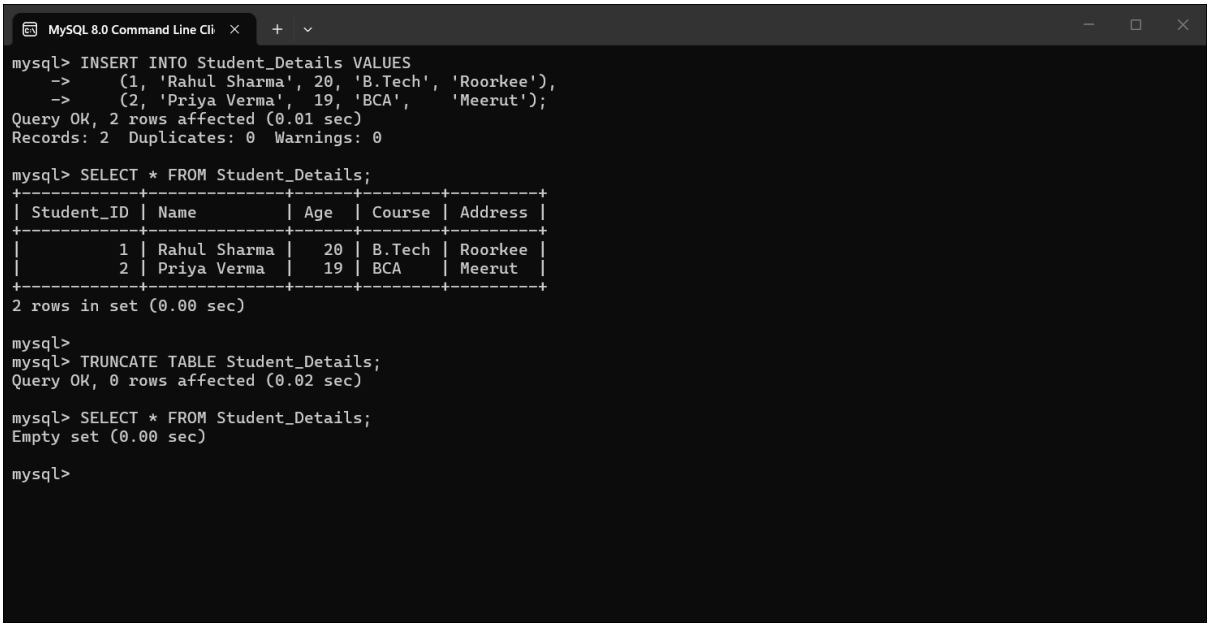
20. Drop the Phone column from the Student table.



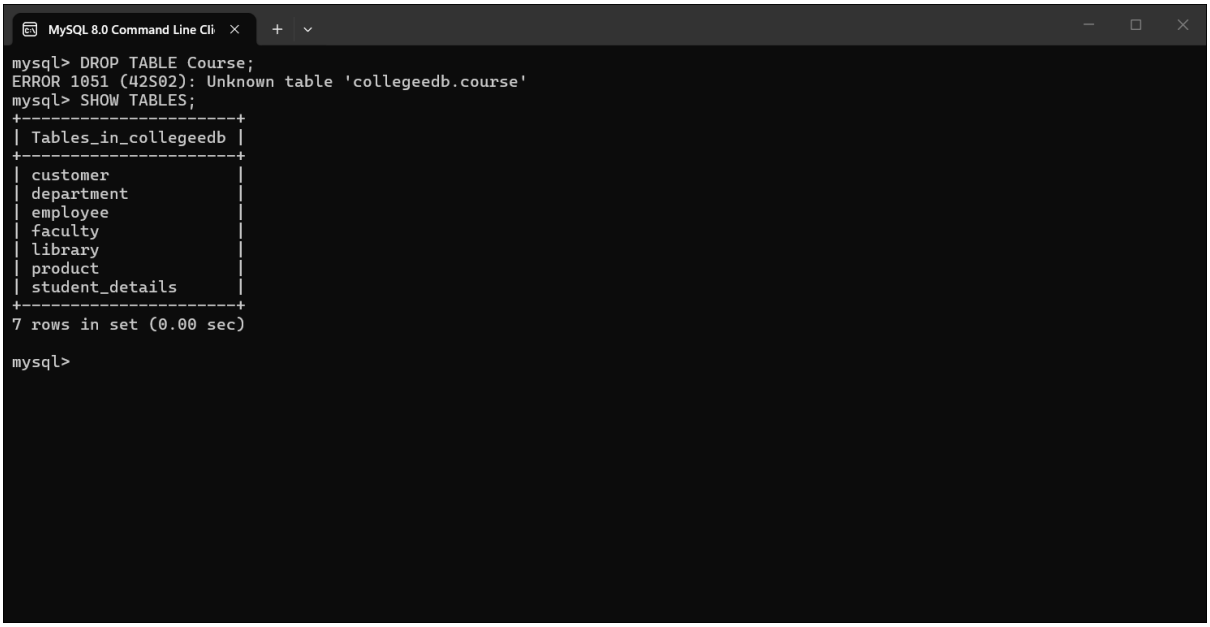
21. Drop the Email column from the Student table.



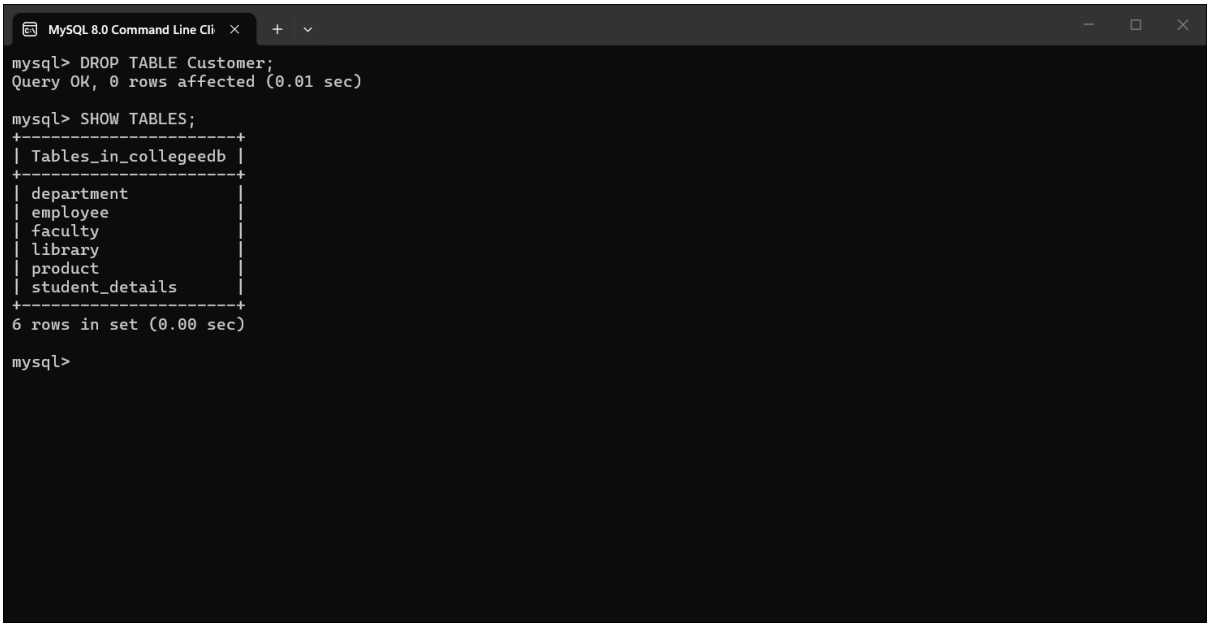
22. Remove all records from a table using TRUNCATE.



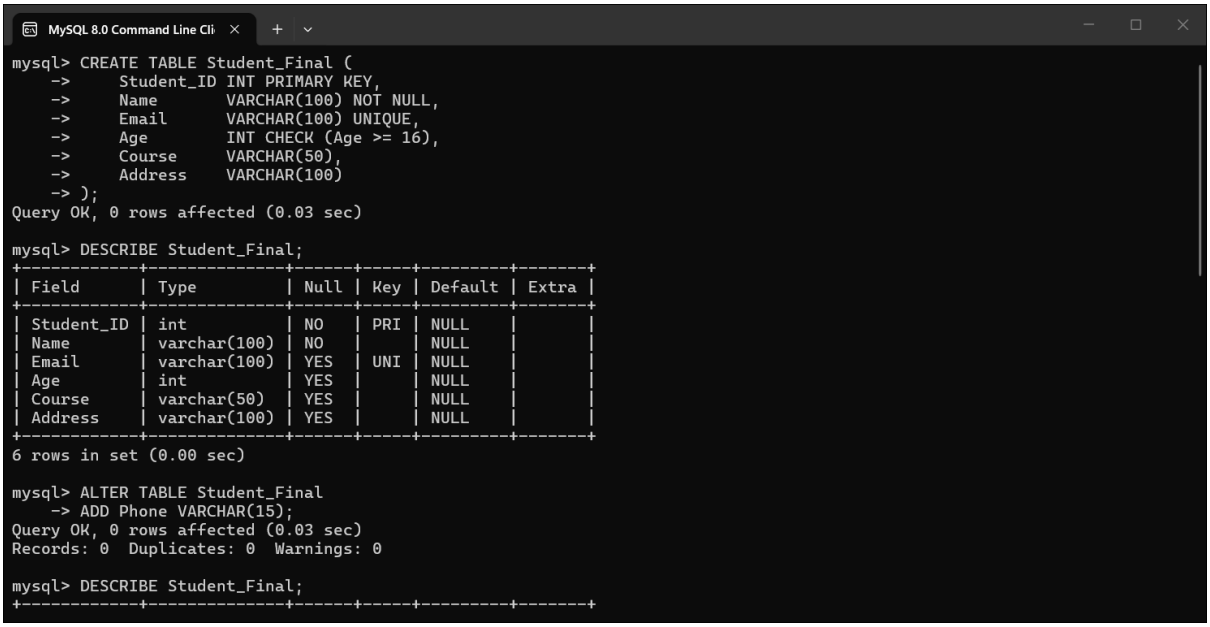
23. Drop the Course table from the database.

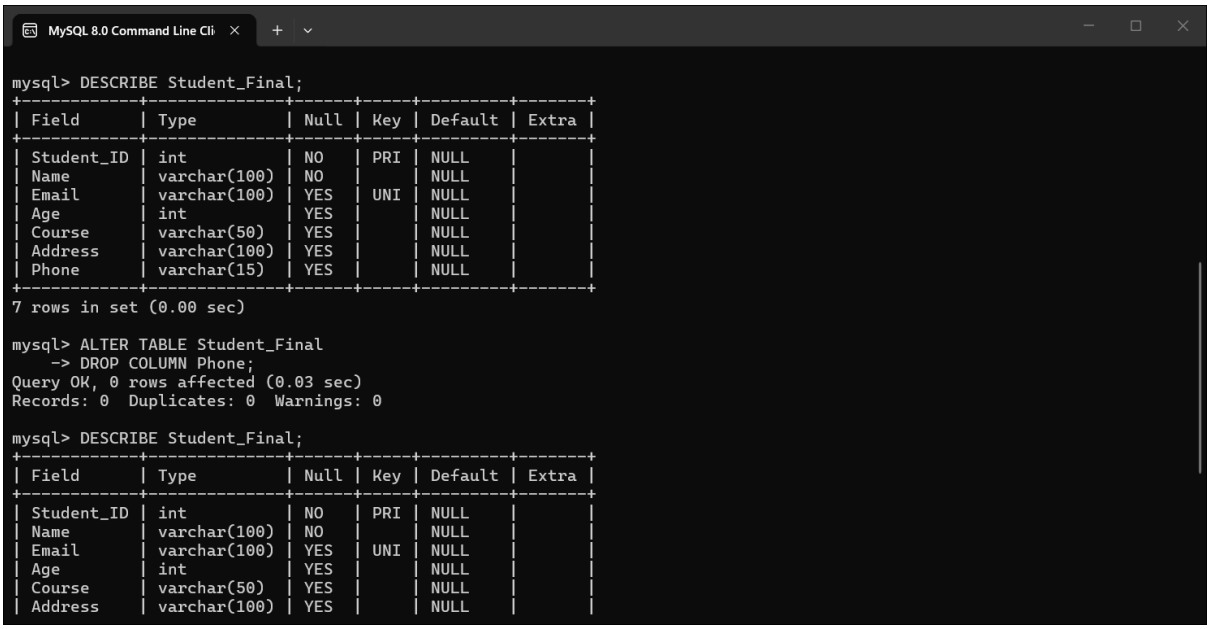


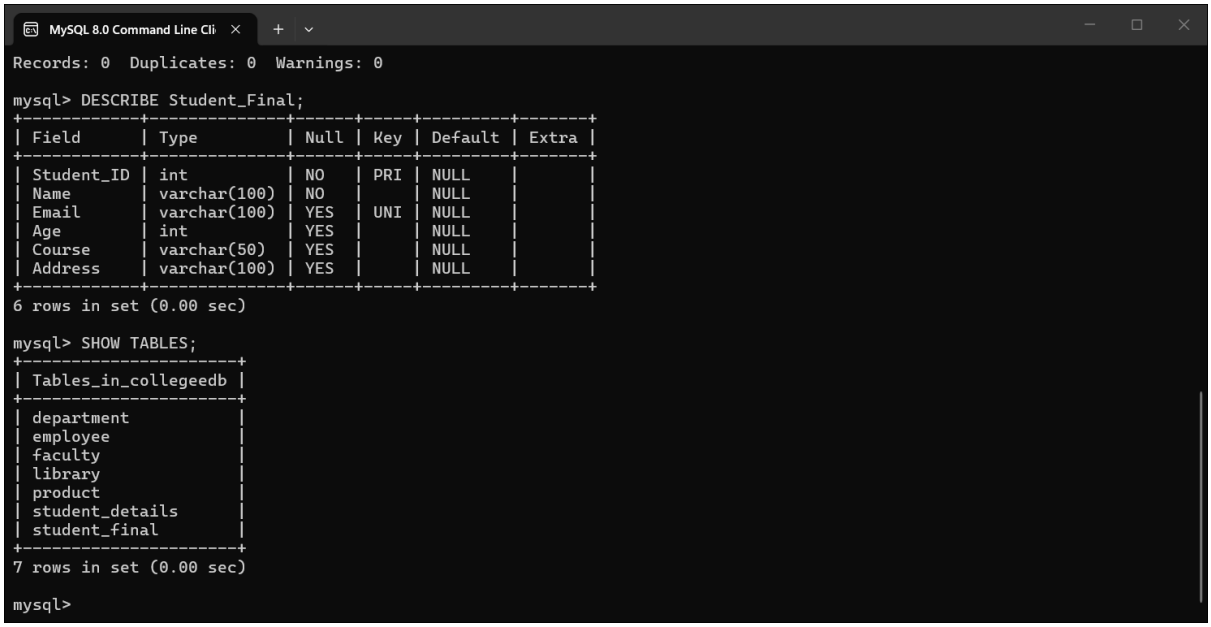
24. Drop the Customer table from the database.



25. Complete Student Table with Constraints


MySQL 8.0 Command Line Cli
mysql> CREATE TABLE Student_Final (
-> Student_ID INT PRIMARY KEY,
-> Name VARCHAR(100) NOT NULL,
-> Email VARCHAR(100) UNIQUE,
-> Age INT CHECK (Age >= 16),
-> Course VARCHAR(50),
-> Address VARCHAR(100)
->);
Query OK, 0 rows affected (0.03 sec)
mysql> DESCRIBE Student_Final;
+------------+--------------+------+-----+---------+-------+
| Field | Type | Null | Key | Default | Extra |
+------------+--------------+------+-----+---------+-------+
Student_ID	int	NO	PRI	NULL	
Name	varchar(100)	NO		NULL	
Email	varchar(100)	YES	UNI	NULL	
Age	int	YES		NULL	
Course	varchar(50)	YES		NULL	
Address	varchar(100)	YES		NULL	
+------------+--------------+------+-----+---------+-------+
6 rows in set (0.00 sec)
mysql> ALTER TABLE Student_Final
-> ADD Phone VARCHAR(15);
Query OK, 0 rows affected (0.03 sec)
Records: 0 Duplicates: 0 Warnings: 0
mysql> DESCRIBE Student_Final;
+------------+--------------+------+-----+---------+-------+


MySQL 8.0 Command Line Cli
mysql> DESCRIBE Student_Final;
+------------+--------------+------+-----+---------+-------+
| Field | Type | Null | Key | Default | Extra |
+------------+--------------+------+-----+---------+-------+
Student_ID	int	NO	PRI	NULL	
Name	varchar(100)	NO		NULL	
Email	varchar(100)	YES	UNI	NULL	
Age	int	YES		NULL	
Course	varchar(50)	YES		NULL	
Address	varchar(100)	YES		NULL	
Phone	varchar(15)	YES		NULL	
+------------+--------------+------+-----+---------+-------+
7 rows in set (0.00 sec)
mysql> ALTER TABLE Student_Final
-> DROP COLUMN Phone;
Query OK, 0 rows affected (0.03 sec)
Records: 0 Duplicates: 0 Warnings: 0
mysql> DESCRIBE Student_Final;
+------------+--------------+------+-----+---------+-------+
| Field | Type | Null | Key | Default | Extra |
+------------+--------------+------+-----+---------+-------+
Student_ID	int	NO	PRI	NULL	
Name	varchar(100)	NO		NULL	
Email	varchar(100)	YES	UNI	NULL	
Age	int	YES		NULL	
Course	varchar(50)	YES		NULL	
Address	varchar(100)	YES		NULL	


MySQL 8.0 Command Line Cli
Records: 0 Duplicates: 0 Warnings: 0
mysql> DESCRIBE Student_Final;
+------------+--------------+------+-----+---------+-------+
| Field | Type | Null | Key | Default | Extra |
+------------+--------------+------+-----+---------+-------+
Student_ID	int	NO	PRI	NULL	
Name	varchar(100)	NO		NULL	
Email	varchar(100)	YES	UNI	NULL	
Age	int	YES		NULL	
Course	varchar(50)	YES		NULL	
Address	varchar(100)	YES		NULL	
+------------+--------------+------+-----+---------+-------+
6 rows in set (0.00 sec)
mysql> SHOW TABLES;
+----------------------+
| Tables_in_collegeedb |
+----------------------+
| department |
| employee |
| faculty |
| library |
| product |
| student_details |
| student_final |
+----------------------+
7 rows in set (0.00 sec)
mysql>